Quantum Measurements as Minkowski Microstates

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Abstract

We show how quantum state reduction naturally plays the role of a microstate in thermodynamic approaches to gravity and develop an information-theoretic language to explore the connection with a concrete construction. Eliminating temperature between Landauer's principle and the Unruh effect yields a purely kinematic acceleration-dependent bound on the energy of information-carrying quanta. Motivated by this, we construct a helicity-anticorrelated Bell state propagating in flat Minkowski space, emitted from a common causal past, that drives the Unruh response in both Rindler wedges. We briefly discuss how proper time can be measured as a property of matter, and then connect the matter action, event by event, to a Gibbons–Hawking–York boundary term and a horizon area increment. Using a pp-wave cut-and-paste surgery, we show how space is dynamically advanced along the horizons. This newly generated geometry carries a concrete label tied to the entangled helicity, which is subsequently observed across the causal horizons, allowing for the measurement of space in geometric and informational terms simultaneously.

1 Introduction

Jacobson derives the Einstein equations from entropy and temperature on local Rindler horizons [1]; Verlinde derives gravitational forces from entropy gradients on holographic screens [2]. Both frameworks operate at the level of bulk thermodynamics and leave open a fundamental question: *what physical process underlies each individual entropy increment?* We will show how the answer can be quantum measurement by developing an information-theoretic language that connects the non-thermal microstates of [3], which can drive the Unruh effect, to current literature.

Before we dig in though, we first briefly outline our high level argument about how quantum measurement naturally connects to gravity. The general setup is pictured in the diagram in Figure 1, starting from Quantum State Reduction and following the solid arrows. Quantum state reduction always happens due to an observation. Any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. And then the equivalence principle identifies this impulse with a local spacetime geometry.[1]

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[1] The dashed arrow in the Figure 1 connects to the Diósi–Penrose postulate [4, 5], which proposes that gravitational effects cause state reduction; the present paper approaches the relationship from the other direction, asking what spacetime geometry a measurement event produces.

$$
\begin{tikzcd}
\text{Spacetime Geometry} \arrow[r, dashed, "\text{spatial bit } \S6"] & \text{Quantum State Reduction} \arrow[d, "\text{Bayesian update } \S3"] \\
\text{Acceleration} \arrow[u, "\text{equivalence principle } \S5"'] \arrow[ur, phantom, "\circlearrowright\ \S4"] & \text{Observation} \arrow[l, "\text{energy–momentum transfer } \S2"]
\end{tikzcd}
$$

Figure 1: $A \xrightarrow{x} B$ means A physically realizes B via x. Solid arrows are established in this paper. The dashed arrow holds for the specific example constructed in Section 6.

1.1 Roadmap

The paper develops each arrow of Figure 1 in turn, unified by a single physical unit: a communication channel between massive observers, mediated by a massless quantum, developed in Section 4.

Section 2 develops the **energy–momentum transfer** arrow: substituting the Unruh temperature into Landauer's principle eliminates temperature entirely, yielding a purely kinematic bound on the minimum energy a carrier must have to register information under acceleration, derived from the irreversibility of the PVM projection at the Rindler horizon. The thrust quantum, in our case a photon carrying the impulse, is the physical carrier of this bound.

Section 3 develops the **Bayesian update** arrow: the POVM-to-PVM transition selects a single non-thermal microstate from the Minkowski vacuum, a helicity-anticorrelated Bell state between right and left Rindler wedges, enforced by their common causal ancestry in the past wedge.

Section 4 is the conceptual core, developing the information-theoretic framework that unifies all three arrows. Matter accumulates a record of communications along its worldline; the horizon is the surface at which quantum entanglement is converted into classical records in matter memory; and the Planck area ℓ_P^2 is the conversion factor between matter action and horizon area.

Section 5 develops the **equivalence principle** arrow: the absorbed photon sources an Aichelburg–Sexl pp-wave at the Rindler horizon, the Raychaudhuri equation gives a discrete area increment $\Delta A = 8\pi G\hbar\omega$, and the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is energy conservation written simultaneously in matter and geometric language.

Section 6 shows how the entangled helicity results in a spatial bit: a measurement of space in geometric and informational terms simultaneously, grounding the top arrow of Figure 1 in a concrete construction.

Several thermodynamic identities recovered along the way, the area law, the Bekenstein bound, the Unruh spectrum, are well established. What is new is locating them microstate by microstate in a single physical event, rather than recovering them only as a bulk average.

2 Observation and Acceleration

The elimination of temperature in this section conceptually and physically leads the way to Section 3, where we construct concrete non-thermal microstates that drive the Unruh response, and the later sections that connect our microstate to the broader literature. It allows us to work in the non-thermal global Minkowski description, where thermality is a derived consequence of restricting to one wedge [6, 7] rather than a primitive.

Section 2.1 derives the Landauer–Unruh bound as an exact kinematic result: substituting the Unruh temperature into Landauer's principle eliminates temperature entirely, leaving a purely kinematic inequality depending only on acceleration, which the construction of Sections 3–5 inherits directly. Section 2.2 identifies the physical carrier of this bound as a thrust quantum and connects the resulting momentum transfer to Verlinde's entropic gravity program. Verlinde-style heuristics recovering standard black hole thermodynamics from this bound are collected in Appendix B.

2.1 The Landauer–Unruh Relation

Landauer's principle [8] establishes $k_B T \, \Delta H$ as the minimum energy dissipated to the environment when a logically irreversible operation[2] reduces the state space by entropy ΔH nats: the operation cannot be undone without this dissipation, independently of how slowly or carefully it is performed:

$$\Delta E \geq k_B T \, \Delta H. \tag{1}$$

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta H.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a temperature-independent lower bound on the carrier energy per nat of information irreversibly registered under proper acceleration a; acceleration sets the minimum energy scale of the carrier.

The connection between Landauer's principle and black hole thermodynamics has been noted previously. Cortês and Liddle [9] showed that Hawking evaporation saturates the Landauer bound at the Hawking temperature, what they term "black hole saturation of the Landauer bound", implying that an evaporating black hole loses information as efficiently as possible. The channel capacity saturation of Section 4.3 supplies the microstate-level account of why this is so: each communication event utilizes the Bremermann capacity exactly, sample by sample. Alvim and Céleri [10] employ similar ingredients, a massless field, a qubit memory, and a channel capacity, to derive a bound on the energy cost of information acquisition via a heat engine argument; their bound depends on the channel capacity and the spacetime metric but temperature is not eliminated and the analysis is restricted to flat spacetime, with horizons identified as the natural next setting [10]. The present bound (3) differs from both in three respects: temperature is eliminated entirely by substituting the Unruh relation, yielding a purely kinematic bound depending only on acceleration; the bound is applied to individual carrier quanta rather than to bulk evaporation or coupling constants; and the microstates are constructed explicitly in Sections 3–5 rather than recovered as bulk averages. The Carnot efficiency bound for a mirror accelerated by photon reflection and the Unruh–Landauer bound (3) are the same inequality approached from opposite directions; this equivalence and its connection to the Geroch process are developed in future work.

[2]The source $\mathcal{P}$, developed in Section 3, prepares a Bell state and emits one photon into each Rindler wedge. The Bell state itself is pure; thermality enters when $\mathcal{R}$ traces out the inaccessible partner in the left wedge, reducing the global pure state to a mixed state over helicities. It is this tracing out, the irreversible loss of the which-wedge information to the environment, that constitutes the logically irreversible operation to which Landauer's principle applies.

This Landauer–Unruh bound is saturated for a photon at one bit when the wavelength equals the horizon distance up to a factor of $4\pi^2/\ln 2$:

$$\lambda_{\mathrm{sat}} = \frac{4\pi^2}{\ln 2} \cdot \frac{c^2}{a} \sim d_{\mathcal{H}}, \tag{4}$$

where $d_{\mathcal{H}} = c^2/a$ is the proper distance to the Rindler horizon. The saturating photon is the fundamental mode of the causal diamond defined by the observer and the horizon, with wavelength commensurate with the horizon distance $d_{\mathcal{H}}$: long enough to remain within the horizon scale and short enough to carry exactly one bit rather than being lost below the noise floor. This identifies the horizon distance as the natural length scale of a single-bit measurement event, and the Rindler horizon as the surface at which the Landauer bound is geometrically realized.

For realistic accelerations the saturation wavelength is enormous, at Earth surface gravity $a = 9.8\ \mathrm{m\,s^{-2}}$, $\lambda_{\mathrm{sat}} \approx 55$ light years, so the single-bit bound is practically always satisfied for terrestrial observers, and is only approached near astrophysical or Planck-scale accelerations.

2.2 Momentum Transfer and Acceleration

The energy transfer ΔE implied by the Landauer–Unruh relation is carried by a physical quantum, matter or radiation, mediating the measurement interaction as a *thrust quantum*. For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{5}$$

This is closely related to Verlinde's entropic gravity [2]: Verlinde identifies holographic thrust on information screens with the origin of inertia itself, and Newton's second law appears in both, Verlinde uses it to identify the entropic force with inertia, while here it gives the photon absorption rate $\dot{N}(a)$ of equation (6) directly. The present approach differs in grounding each entropy increment in a discrete measurement event, the measurement impulse playing the role of Verlinde's holographic thrust, rather than in a coarse-grained thermodynamic gradient.

The Landauer–Unruh bound is saturated not at the level of an individual photon but at the level of the source rate[3]. The acceleration a does not fix the frequency ω of each selected quantum; rather, it determines the rate at which the source selects entangled pairs in order to implement that acceleration, with the total power $\dot{N}\hbar\omega$ matching the Unruh thermal flux (as demonstrated at saturation in Section 4.3). Each individual photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome, one bit in the sense of one binary alternative carrying $\ln 2$ nats, and these are paired by the construction independently of a. The Landauer bound then operates as a consistency condition on the ensemble: the minimum carrier energy per bit that clears the Unruh noise floor, summed over the acquisition rate, equals the power required to sustain the acceleration.

Applying Newton's second law to the momentum impulse $\hbar\omega/c$ per absorbed photon gives the interaction rate directly for a detector of mass m at rest:

$$a = \frac{\dot{N}(a)\,\hbar\omega}{mc}, \qquad \dot{N}(a) = \frac{mca}{\hbar\omega}. \tag{6}$$

Two consequences of this rate and the bound (3) are developed later: the ensemble of individual non-thermal microstates selected at rate $\dot{N}(a)$ is shown in Section 4.3 to saturate the channel capacity exactly, providing a dynamical account of how thermality emerges from coarse-graining over individual measurement events; and the bound (3) is shown in Appendix B.3 to coincide with the Bekenstein universal entropy bound at the horizon condition $a = c^2/R$, identifying the two as the same inequality in acceleration-frame and position-space descriptions respectively.

[3] Here we discuss a constant rate with a fixed frequency, but many situations are possible.

3 Bilocal Microstate Construction

From now on we use natural units $\hbar = c = 1$ unless otherwise specified; factors of $\hbar$ and c are restored where they aid physical interpretation. G is retained explicitly throughout, so that the Planck area $\ell_P^2 = \hbar G/c^3$ remains a visible conversion factor between matter and geometric descriptions.

The arguments of Section 2 motivate a relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. In [3] this logic was inverted for a scalar field: a source placed in the past injects non-thermal entangled excitations into the field, and the observer's accelerated response was shown to be driveable by these source-induced excitations. The thermal/non-thermal transition was seen to be governed by the Stokes phenomenon with asymptotics of parabolic cylinder functions, the non-thermal limit was recovered by projecting onto a specific frequency ω using a projective measurement (PVM) whose operator structure is further developed in Section 3.2. We extend that construction here to the spin-1 electromagnetic field, upgrading from scalar to helicity-carrying Bell states and making the single-bit structure of each measurement event explicit.

3.1 A Source and Two Observers

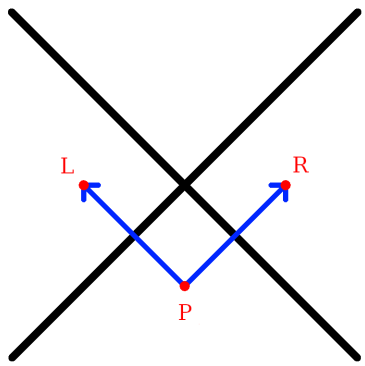

Figure 2: A source in the past, $\mathcal{P}$, emits an entangled photon pair that accelerates $\mathcal{R}$ in the right Rindler wedge and $\mathcal{L}$ in the left Rindler wedge. The two null directions of propagation define the null coordinates $u = t-z+1$ and $v = t+z+1$, with the right-moving photon propagating along $u = 0$ and the left-moving photon along $v = 0$; these are the z^- and z^+ axes of the double-null coordinate system of Section 3.4.

Three massive bodies are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named $\mathcal{R}$ – *Right*, $\mathcal{L}$ – *Left*, and $\mathcal{P}$ – *Past* respectively. $\mathcal{P}$ serves as the common causal ancestor of the accelerations of $\mathcal{R}$ and $\mathcal{L}$ (see Figure 2). $\mathcal{P}$ emits an entangled pair of spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k \equiv |k|$, the frequency associated to momentum k for a massless particle in natural units. The source exhibits no net back-reaction in either linear or angular momentum[4]. The two photons

[4]The momenta are perfectly balanced by construction: the two photons carry spatial momenta $+k\hat{z}$ and $-k\hat{z}$ respectively, so the total linear momentum vanishes, $k + (-k) = 0$. The angular momentum vanishes term by term: in each helicity assignment $(\lambda_R, \lambda_L) = (+1, -1)$ and $(-1, +1)$, the total helicity is $\lambda_R + \lambda_L = (+1) + (-1) = 0$ and $(-1) + (+1) = 0$ respectively, and the symmetric Bell state superposition (18) preserves this since both terms contribute zero independently

propagate into the left and right Rindler wedges, where $\mathcal{L}$ and $\mathcal{R}$ detect them respectively via absorption. The momentum impulse accelerates each detector in its respective Rindler wedge; each observer thereby registers a helicity eigenvalue and acquires one classical bit of information. The emission event at $\mathcal{P}$ naturally defines null coordinates $u = t - z + 1$ and $v = t + z + 1$, centered so that $\mathcal{P}$ lies at $u = v = 0$, the right-moving photon propagates along $u = 0$, and the left-moving photon propagates along $v = 0$.

The three-party structure $\mathcal{P}$–$\mathcal{R}$–$\mathcal{L}$ realizes the thermofield double in flat Minkowski space: $\mathcal{R}$ sees a thermal state because $\mathcal{L}$ exists and is causally inaccessible, with the correlations carried by the off-diagonal entries of the Bell state rather than by any exotic quantum gravity effect. The classical bit recorded by $\mathcal{R}$ must be anticorrelated with that recorded by $\mathcal{L}$ in the same basis; a future observer $\mathcal{F}$ with access to both records recovers the full Bell state: the correlations are classical records in matter memory, accessible without any gravitational reconstruction procedure.

3.2 PVM Localization and the Elimination of Coarse-Graining

The Unruh mode plays a double role in this construction. As a QFT object it is a coefficient in a mode expansion, a solution to the wave equation analytically continued from the Rindler wedge into all of Minkowski space. But once $\mathcal{P}$ is specified as a real physical source via the squeezing interaction, equation (15), the mode function acquires a second interpretation: contracted against the detector's response function via the Born rule, it gives the probability amplitude for where $\mathcal{R}$ is likely to detect the emitted quantum.

This is the inversion of the standard Unruh–DeWitt logic. Rather than coupling a detector to a pre-existing acceleration and reading off a thermal distribution, we front-load the logic at $\mathcal{P}$. In the standard Unruh–DeWitt treatment [11, 12], the detector response integrates over all proper time $\tau \in (-\infty, \infty)$; since each proper time corresponds to a different local redshift along the accelerated worldline, this is simultaneously an average over all redshifted Rindler frequencies, and their joint product is the Planck distribution. Here instead the source fixes the frequency ω_k and the mode function specifies the probability amplitude for detection, collapsing both coarse-grainings simultaneously by fixing u and ω_k together.

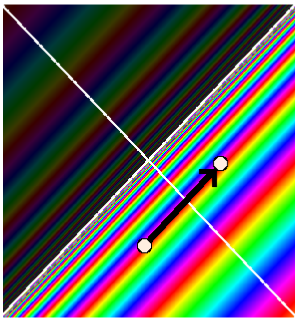

Figure 3: The points $\mathcal{P}$ and $\mathcal{R}$ are marked with white dots, connected by the null geodesic (black arrow). The rainbow coloring shows the Unruh mode phase structure across Minkowski space. A PVM projection onto this null line selects a single frequency from the thermal superposition, replacing the full delocalized mode with the localized non-thermal microstate of equation (18).

Longitudinal localization. The primary localization is in the longitudinal (t, z) plane, which is the setting of Figure 3 and of the parabolic cylinder function analysis of [3]. In that work, a Minkowski mode φ_q^* multiplied pointwise by a Gaussian shaping function $N_{\mu,\sigma}$ of width σ in (t, z), then paired via the Klein–Gordon inner product with a Rindler mode r_k, yields an overlap expressible in terms of parabolic cylinder functions $D_\nu(-\zeta)$. The Gaussian could equivalently have been applied to the Rindler side; what matters is that σ controls the degree of longitudinal localization of the source-detector pair. The Stokes phenomenon, see Figure 4, in the asymptotics of $D_\nu(-\zeta)$ as σ varies drives a transition between two physically distinct regimes: $\sigma \to \infty$ gives the thermal regime where the $e^{-\frac{1}{4}\zeta^2}$ branch dominates, and $\sigma \to 0$ gives the localized non-thermal regime where the $e^{+\frac{1}{4}\zeta^2}$ branch dominates, with the Stokes line at $\arg\zeta = \pi/4$ marking the boundary between them.

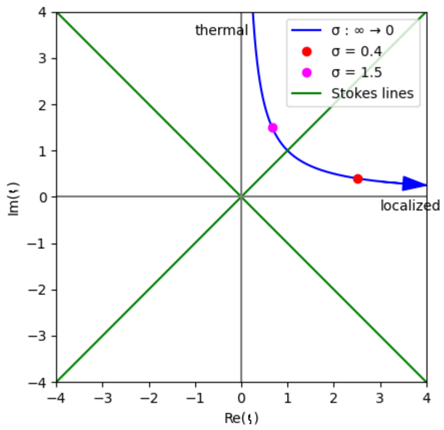

Figure 4: Trajectory of $\zeta = iq\sigma + \mu/\sigma$ in the complex plane as σ interpolates between the two limits of the Kraus family (7). The thermal limit $\sigma \to \infty$ corresponds to ζ on the positive imaginary axis; the localized PVM limit $\sigma \to 0$ corresponds to ζ on the positive real axis. The green Stokes lines at $\arg\zeta = \pi/4$ mark the boundary between the two asymptotic regimes of the parabolic cylinder function $D_\nu(-\zeta)$, and therefore the boundary between the POVM and PVM descriptions of the longitudinal localization. Figure from [3].

In the present paper the same Gaussian shaping function is reread as a Kraus operator acting on the quantum state,

$$K_\sigma = \int e^{-u^2/2\sigma^2}\, |u\rangle\langle u|\, du, \tag{7}$$

where $u = t - z + 1$ is the null coordinate[5] defined in Section 5, and $|u\rangle$ are generalized null-position eigenstates in the sense of a rigged Hilbert space, satisfying $\int |u\rangle\langle u|\, du = \mathbf{1}$ distributionally. The Gaussian envelope $e^{-u^2/2\sigma^2}$ renders K_σ a bounded operator, and the completeness relation ensures that $K_\sigma^\dagger K_\sigma + (K_\sigma^\perp)^\dagger K_\sigma^\perp = \mathbf{1}$ as required for a valid POVM. The two readings of K_σ are consistent because the Klein–Gordon inner product integrand, which in [3] computed the overlap between a Gaussian-shaped source mode and a Rindler mode, is the Born-rule probability amplitude for the same measurement event: what was a preparation in the source picture is a projection in the measurement picture, and the Gaussian width σ controls the sharpness of that projection.

The operator K_σ together with its complement

$$K_\sigma^\perp = \int \sqrt{1 - e^{-u^2/\sigma^2}}\, |u\rangle\langle u|\, du, \tag{8}$$

satisfying $K_\sigma^\dagger K_\sigma + (K_\sigma^\perp)^\dagger K_\sigma^\perp = \mathbf{1}$ by construction, defines a one-parameter POVM family. Physically, finite σ corresponds to a source delocalized over a range of null positions: no

[5]We use ζ for the Stokes variable of [3] to avoid confusion with the longitudinal spatial coordinate z.

single null geodesic from $\mathcal{P}$ to $\mathcal{R}$ is selected, and the detector sees a mixture because it has no information about which geodesic the photon traveled. The parameter σ therefore quantifies ignorance about the source location, and thermality is the consequence of that ignorance. The physical content of this family is revealed by its asymptotic behavior: the POVM family (7) is the measurement-picture object whose Stokes transition the companion paper computes.

The distinction between preparation and measurement is operationally sharp [13]: *mode shaping* is a preparation that modifies the physical field; *PVM projection* is a measurement that conditions the observer's description on causal data without touching the ontic state. The Gaussian width σ of the Kraus family (7) admits both readings simultaneously: as a source localization parameter in the preparation sense of [3], and as an epistemic resolution parameter in the measurement sense here, with the Stokes line marking the boundary at which the two readings give qualitatively different physics. The direction of explanation is thereby reversed: the emission event at $\mathcal{P}$ is primary, and the geometry is the ontic output of the subsequent measurement events, while the assignment of that output to a particular observer's microstate record remains epistemic.

Summarizing, the relationship to the companion paper [3] is one of conceptual upgrade. That paper computed a Klein–Gordon overlap integral between a Gaussian-shaped source mode and a Rindler mode, obtaining a parabolic cylinder function whose Stokes transition marks the boundary between thermal and non-thermal regimes. The present paper provides the physical interpretation: the Gaussian-times-mode object is a POVM family parameterized by σ, acting on the quantum state, and the POVM/PVM transition is the physical content. The measurement language is what makes the Stokes transition physically meaningful.

Transverse localization. The transverse (x, y) directions require a separate and independent localization: a Rindler plane wave at frequency ω is a coherent superposition over all possible source-detector pairs $(\mathcal{P}', \mathcal{R}')$ connected by null geodesics at that frequency, uniform in the transverse plane, so specifying $\mathcal{P}$ and $\mathcal{R}$ performs a PVM projection in the transverse directions as well. The longitudinal and transverse localizations decouple because the wave equation in double-null coordinates separates: the u-dependence and the (x, y)-dependence of the mode function enter independently, so the two Kraus projections can be applied in either order. The transverse shaping is controlled by an independent Kraus family

$$K_{\sigma_\perp} = \int e^{-r^2/2\sigma_\perp^2} \, |\mathbf{x}_\perp\rangle\langle\mathbf{x}_\perp| \, d^2x_\perp, \tag{9}$$

the direct transverse analogue of (7). The total horizon area increment $\Delta A = 8\pi G\hbar\omega$ is independent of $\sigma_\perp$ by energy conservation, as the Raychaudhuri calculation of Section 5 shows: only the total null energy $\int T_{uu}\, du\, d^2x_\perp = \hbar\omega$ enters, not the transverse distribution. We work in the localized limit $\sigma_\perp \to 0$, which recovers the Aichelburg–Sexl pp-wave developed in Section 5.

The area increment $\Delta A = 8\pi G\hbar\omega$ and the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ in Section 5, which serve as the geometric consistency checks of the framework, are independent of $\sigma_\perp$, fixed by energy conservation alone.

The microstate. The result of both projections is an individually non-thermal microstate carrying a definite Rindler frequency ω and a definite longitudinal position on the null geodesic from $\mathcal{P}$ to $\mathcal{R}$. Thermality is not a property of any single microstate but of the ensemble; it emerges when one models the detector response within a wedge as a sum over all Rindler modes, as in the standard Unruh–DeWitt treatment [11, 12].

Throughout this paper the PVM limit is understood as $\sigma \to 0^+$: the Kraus operator (7) is not normalizable at exactly $\sigma = 0$, and the AS shift $\Delta v \propto -\ln r^2$ diverges on axis, both regularized by any $\sigma > 0$. The area increment $\Delta A = 8\pi G\hbar\omega$ is independent of σ by energy conservation. The parameter σ is simultaneously the longitudinal Kraus width and the transverse profile width.

3.3 Helicity Structure and PCT Symmetry

We now upgrade this construction from a spin-0 scalar field in [3] to the spin-1 electromagnetic field with a superposition of helicity pairs in a Bell state, introducing one unknown binary degree of freedom, the helicity outcome, undetermined until measured, which is the single bit responsible for the remaining thermality in the ensemble. The photon carries one quantum of action $\hbar$ and energy $E = \hbar\omega$, identifying the thrust quantum of Section 2 with a single photon whose polarization outcome, upon detection, is the classical bit acquired by each observer.

The Bell state is the natural state in this momentum sector invariant under the PCT symmetry between wedges, implemented by the Bisognano–Wichmann [7] modular conjugation J, which maps the right-wedge algebra to the left-wedge algebra and simultaneously flips $\lambda \to -\lambda$; the anticorrelated helicity assignment is the direct consequence. The common algebraic origin of this anticorrelation in the Lorentz algebra decomposition $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$ is developed in Appendix A, and the geometric signature the photons imprint on the horizon is developed in Section 5.[6]

3.4 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector $k_R^\mu = (\omega, 0, 0, k)$ and the left-moving photon has wave-vector $k_L^\mu = (\omega, 0, 0, -k)$, with equal and opposite spatial momenta. These two null directions are precisely the z^+ and z^- axes of double-null coordinates

$$z^\pm = \frac{1}{\sqrt{2}}(t \pm z) = \frac{1}{\sqrt{2}}(u - 1 \pm (v - 1)), \tag{10}$$

with transverse coordinates $\mathbf{z}_\perp = (x, y)$. Note that $z^\pm$ are centered on the Minkowski origin $(t, z) = (0, 0)$ rather than on $\mathcal{P}$; the coordinates u, v defined in Section 3 are the $\mathcal{P}$-centered versions and are used throughout the remainder of the paper. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(z) = 0, \qquad A_-(z) = 0 \tag{11}$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^x, A^y)$. This gauge respects the PCT symmetry that swaps the two wedges ($z^+ \leftrightarrow z^-$) simultaneously with ($R \leftrightarrow L$), and therefore treats both observers symmetrically by construction. The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon^i_{(\lambda)}$ with $\lambda = \pm 1$.

The lack of manifest four-dimensional covariance in equation (11) is a consequence of aligning the gauge with the pair of physical null directions selected by the source. The full Lorentz group is broken to the little group preserving the pair $\{k_R^\mu, k_L^\mu\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice: the helicity eigenvalues are Poincaré-covariant objects independent of how the gauge is fixed.

[6] The helicity label λ enters the longitudinal mode function as a spectator index and does not affect the u-dependence of the Kraus operator (7); the Stokes transition analysis of [3], which is carried out in 1+1 dimensions, therefore applies independently to each helicity sector without modification. The extension to 3+1 dimensions introduces the transverse coordinates (x, y) as an independent sector: because the wave equation in double-null coordinates separates, the longitudinal Stokes analysis is unaffected by the transverse directions, which are handled separately by the transverse Kraus family equation (9).

3.5 The Bilocal Source

In double-null gauge the electromagnetic field is characterized by its transverse components alone. The bilocal driving source is symmetrized over both helicity assignments,

$$\mathcal{J}^{ij}(x,y) = \left(\epsilon^{i*}_{(-1)}\,\epsilon^{j}_{(+1)} + \epsilon^{i*}_{(+1)}\,\epsilon^{j}_{(-1)}\right) f_L(x)\, f_R(y), \tag{12}$$

where f_L and f_R are normalized Rindler mode profiles supported on the left and right wedges respectively, and $\epsilon^i_{(\lambda)}$ are the circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}. \tag{13}$$

The two terms in (12) are the two PCT-related helicity assignments established in Section 3.4: the modular conjugation J maps the first to the second and vice versa, and the symmetric superposition is the simplest state respecting this exchange symmetry with total angular momentum zero in each term independently.

The bilocal driving Lagrangian is

$$\mathcal{L}_{\text{sourced}} = \mathcal{L}_{\text{free}} + \frac{1}{2}\int \mathcal{J}^{ij}(x,y)\, A_i^+(x)\, A_j^+(y)\, d^4x\, d^4y + \text{h.c.}, \tag{14}$$

where A_i^+ denotes the positive-frequency part of the transverse gauge field. Smearing f_L and f_R with A_i^+ projects them onto their positive-frequency parts, which are the Unruh mode profiles μ_L and μ_R respectively. The creation operators $c^{L\dagger}_{-k,(\lambda)}$ and $c^{R\dagger}_{k,(\lambda)}$ are the ladder operators associated with these Unruh modes: the right-moving photon of energy $\hbar\omega$ is deposited in the right wedge and the left-moving photon in the left wedge as a consequence of this projection, not as an additional assumption about the source support. The operator content of this selection is

$$c^{L\dagger}_{-k,(-1)}\, c^{R\dagger}_{k,(+1)} \;+\; c^{L\dagger}_{-k,(+1)}\, c^{R\dagger}_{k,(-1)} \;+\; \text{h.c.} \tag{15}$$

The bilocal source selects the $n = 1$ term in the vacuum decomposition (16) at fixed ω and fixed helicity pair $(\lambda, -\lambda)$ via the PVM projection of Section 3.2: no perturbative expansion is required, and the selection is prescriptive rather than perturbative. Each photon carries energy $E = \hbar\omega$ and one quantum of action $\hbar$; its polarization, undetermined until measured, is the quantum bit whose outcome constitutes the classical record acquired by each observer upon detection.

3.6 Vacuum Decomposition and Microstate

The Minkowski vacuum thermofield double state in the Rindler basis reads, for each helicity sector,

$$|0\rangle_M^{(\lambda)} = \frac{1}{\cosh\theta_\omega}\sum_{n=0}^{\infty}\tanh^n\theta_\omega\; |n_{\omega,\lambda}\rangle_R\, |n_{\omega,-\lambda}\rangle_L, \tag{16}$$

where $\tanh\theta_\omega = e^{-\pi\omega c/a}$ encodes the Unruh distribution at acceleration a, and the full vacuum is the product over all ω and both helicities:

$$|0\rangle_M = \prod_{\omega,\lambda} |0\rangle_M^{(\lambda)}. \tag{17}$$

The helicity label λ enters as a spectator index throughout: the anticorrelation $\lambda \to -\lambda$ between wedges is preserved at every level of the construction, a direct consequence of the Bisognano–Wichmann modular conjugation established in Section 3.

The bilocal source of Section 3.5 selects the $n = 1$ term in (16) at fixed ω and fixed helicity pair $(\lambda, -\lambda)$ via the PVM projection of Section 3.2. The resulting single microstate is

$$|\Psi_\omega\rangle_M = \frac{1}{\sqrt{2}} \left(|1_{\omega,+1}\rangle_R \, |1_{\omega,-1}\rangle_L + |1_{\omega,-1}\rangle_R \, |1_{\omega,+1}\rangle_L \right), \tag{18}$$

with $\mathcal{R}$'s and $\mathcal{L}$'s helicity outcomes undetermined until measurement, at which point they are perfectly anticorrelated: each observer acquires one classical bit, and the two bits are complements of one another.

4 Matter as a Substrate

We explore a language that realizes our microstate as a building block for emergent spacetime. Specifically, we propose that matter is the fundamental substrate of spacetime information processing by reinterpreting quantum mechanics in terms of communication channels. The following subsections develop the idea:

1. **Matter as a timelike memory** (Section 4.1): Matter accumulates a record of communications along its worldline at the Bremermann sampling rate; proper time is measured as accumulated action $S/\hbar$, with discreteness living in the matter sector via $\hbar$. The Bremermann channel capacity C is a property of the receiver alone, independent of acceleration.
2. **The horizon as a spacelike channel** (Section 4.2): Spatial geometry emerges from communications between $\mathcal{P}$ and $\mathcal{R}$ through a horizon; the Planck area[7] ℓ_P^2 is the conversion factor between matter action and horizon area.
3. **Channel capacity saturation** (Section 4.3): The Bremermann channel capacity C, the Newtonian interaction rate $\dot{N}(a)$, and the Landauer–Unruh noise floor coincide at the horizon, where every Bremermann sample receives exactly one write operation and C is exactly utilized sample by sample.

In this way, space and time can be measured and sampled as properties of matter and interactions. Section 5 will use this language to connect matter action S_{matter} to the Gibbons–Hawking–York boundary term $S_{GHY}|_{\mathcal{H}}$ sample by sample.

Our approach dissolves a fundamental question about quantum gravity: *how can one quantize gravity without directly quantizing spacetime or its curvature degrees of freedom?* Common approaches posit Planck-scale cells [16] or discrete spacetime events [17], but Lorentz invariance is recovered only statistically in such constructions rather than at the level of individual configurations [18]. We instead recover discreteness from matter itself: space and time are properties of matter and are defined only up to the resolution set by the action $S/\hbar$, which is Lorentz invariant by construction.

4.1 Matter as a Timelike Memory

A massive particle carries phase along its worldline at a rate set by its rest energy. The Compton frequency $f = mc^2/h$ is the natural clock rate of a mass m: it is the rate at which the phase $e^{iS/\hbar}$ completes one cycle. This is the content of Bremermann's computational limit [19]: the maximum sampling rate of any physical system with bounded rest energy.

We associate a discrete event with each zero crossing of the phase $e^{iS/\hbar}$ in the path integral and use the information-theoretic term *sample* throughout. This association is

[7] The three fundamental constants c, $\hbar$, and G play distinct roles in this picture. The speed of light c sets the causal propagation speed of signals. The reduced Planck constant $\hbar$ sets the resolution of the matter via the Compton frequency mc^2/h. The gravitational constant G, through the Schwarzschild radius $r_s = 2Gm/c^2$, sets the maximum energy storable in a spatial region without gravitational collapse. Their combination $\ell_P^2 = \hbar G/c^3$ is the Planck area, the minimal spatial region per independent quantum of action

a definition: it identifies the natural discreteness of the matter action with the sampling resolution of the Bremermann limit, and makes no additional physical postulate beyond the path integral itself. The sampling resolution is limited by the Heisenberg uncertainty $\Delta E \Delta t \sim \hbar$, which plays the role of the Nyquist limit for the matter memory: it would require additional energy $\Delta E > mc^2$ to probe the sub-Compton phase. The signal being sampled at the Bremermann rate is the internal quantum state of the particle, its spin, helicity, and other quantum numbers, and the dynamic range per sample is $\ln d$ nats, where d is the dimension of the internal Hilbert space.

Definition 4.1. *The Bremermann channel capacity is the maximum rate at which a massive body can accumulate distinguishable records along its worldline. For a system of mass m with internal Hilbert space dimension d:*

$$C = \underbrace{\frac{mc^2}{h}}_{\text{sampling rate}} \times \underbrace{\ln d}_{\text{dynamic range}} \qquad \textit{(nats per second).} \tag{19}$$

This is a property of the receiver alone, independent of acceleration and of the rate at which communications arrive.

Both sampling rate and dynamic range are extensive under composition:

$$\frac{(m_1 + m_2)c^2}{\hbar} = \frac{m_1 c^2}{\hbar} + \frac{m_2 c^2}{\hbar}, \tag{20}$$

$$\ln(d_1 d_2) = \ln(d_1) + \ln(d_2), \tag{21}$$

consistent with their interpretation as properties of matter that compose naturally under union.

A concrete QFT realization of this framework is the Dirac field in the two-component formalism, where the zig-zag of opposite-helicity Weyl spinors coupled at the Compton scale is precisely the tick-tock mechanism of Appendix A: matter as a timelike memory is the natural QIT language for an established QFT result.

4.2 The Horizon as a Spacelike Channel

The horizon enters the channel picture in a double role. As a *channel*, it is the surface through which causal connection is established: the photon crosses it and that crossing is a communication, converting quantum entanglement between the wedges into a classical record in matter memory. As a *barrier*, it is the surface at which this conversion makes the opposite wedge causally irrelevant to the receiving observer's future evolution. The disconnection between wedges is therefore not a pre-existing feature of the geometry that the photon happens to cross; it is a consequence of the communication itself. Before the crossing $\mathcal{R}$ and $\mathcal{L}$ share a global pure state; after it, $\mathcal{L}$ is inaccessible to $\mathcal{R}$ precisely because the entanglement has been consumed in producing the classical record.

The photon crossing the horizon is a transfer of information and action between the timelike matter memories of $\mathcal{R}$ and $\mathcal{L}$, mediated by the spacelike horizon channel. To that end we define:

Definition 4.2 (Communication)**.** *A communication is a horizon-crossing particle*[8] *that transfers action and information between matter memories.*

In the explicit bilocal construction of Section 3, this takes the following specific form:

(i) a massless quantum of Rindler frequency ω and helicity $\lambda = \pm 1$, selected by the PVM projection of Section 3.2, transferring one quantum of action $\hbar$ across $\mathcal{H}$;

[8] For our microstate this is a photon; Appendix A develops a natural generalization to massive particles.

(ii) the receiving matter memory increasing its Bremermann sampling rate by $\Delta f = \omega/2\pi$;

(iii) the helicity outcome λ written as one classical bit into the receiver's memory; and

(iv) the horizon area incrementing by $\Delta A = 8\pi G\hbar\omega$.

Property (iv) follows from the Raychaudhuri equation applied to the null congruence at $\mathcal{H}$ and is derived in Section 5; it is listed here to make the geometric content of the communication explicit. The restriction to photon carriers and PVM projections is a deliberate scope limitation rather than a fundamental one: the four properties (i)–(iv) define the minimal communication unit for the explicit bilocal construction of Section 3, and whether they extend to massive carriers or POVM measurements is a question the construction motivates but does not address.

Each communication has exactly two irreducible constituents: a transfer of energy $\hbar\omega$ to the receiving matter memory, incrementing its Bremermann sampling rate by $\Delta f = \omega/2\pi$, and a record of one classical bit, the helicity outcome λ. These are not independent: they are the energy and angular momentum of a single photon, two properties of the same physical object. With this definition in hand, the equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ of Section 5.4 is energy conservation at the horizon, written once in matter language and once in geometric language: the GHY boundary term sums $\hbar\omega$ over communications because that is what the term geometrically records, one contribution per event.

4.3 Channel Capacity Saturation

Write operations and the Landauer–Unruh bound. The Bremermann channel capacity C describes the receiver in isolation. Writing into that memory requires a communication: a photon of frequency ω crosses the horizon and is absorbed by $\mathcal{R}$, depositing one quantum of action $\hbar$ and one helicity outcome into matter memory. For this write operation to produce a distinguishable record rather than being lost to the Unruh thermal bath, the photon energy must clear the noise floor at acceleration a. By the Landauer–Unruh bound (3), a successful one-nat write requires $\hbar\omega \geq \hbar a/2\pi c$, following Landauer's argument [8] that any signal must have energy exceeding the thermal noise floor per degree of freedom to constitute a distinguishable record. This condition is naturally expressed as a signal-to-noise ratio, identifying the signal energy as $\hbar\omega$ and the Unruh energy scale $\hbar a/2\pi c$ as the noise floor:

$$\text{SNR} = \frac{\hbar\omega}{\hbar a/2\pi c} = \frac{2\pi c\,\omega}{a}, \tag{22}$$

so that the UL bound is simply SNR ≥ 1 per nat. A write operation succeeds as a distinguishable record if and only if its SNR clears this threshold.

Interaction rate and saturation. The interaction rate $\dot{N}(a)$ of equation (6), derived from Newton's second law applied to $\mathcal{R}$ at rest in Section 3, connects the acceleration a to the Bremermann channel capacity C as follows. The saturation frequency ω_{sat} is the minimum frequency for a successful $\ln d$-nat write against the Unruh noise floor:

$$\omega_{\text{sat}} = \frac{a\ln d}{2\pi c}. \tag{23}$$

Substituting into equation (6):

$$\dot{N}_{\text{sat}} = \frac{m_{\mathcal{R}}ca}{\hbar}\cdot\frac{2\pi c}{a\ln d} = \frac{m_{\mathcal{R}}c^2}{h}\cdot\frac{1}{\ln d} = \frac{f_{\mathcal{R}}}{\ln d}. \tag{24}$$

The throughput at saturation is therefore

$$\dot{N}_{\text{sat}}\cdot\ln d = f_{\mathcal{R}} = C, \tag{25}$$

The capacity is exactly saturated: the Bremermann channel capacity C is utilized sample by sample, with no free parameters and no remainder. This is not a dimensional coincidence but a consequence of Newton's second law, the Landauer–Unruh bound, and the Bremermann limit converging to the same condition at the horizon, three independently motivated constraints that turn out to be the same constraint written in three different languages. Since the Landauer–Unruh bound and the Bekenstein bound are the same inequality at the horizon condition $a = c^2/R$ (Appendix B.3), saturation of C is the information-theoretic expression of both simultaneously.

The saturation condition is not confined to the Rindler setting: as shown in Appendix B, it connects directly to black hole thermodynamics via the Bekenstein–Hawking area law, with the horizon condition $a = c^2/R$ identifying the Landauer–Unruh bound and the Bekenstein universal entropy bound as the same inequality in different frames. The framework therefore touches ground in two independent places: the Dirac field in flat spacetime and black hole thermodynamics in the curved setting.

The SNR at saturation from equation (23) is

$$\mathrm{SNR}_{\mathrm{sat}} = \frac{2\pi c\, \omega_{\mathrm{sat}}}{a} = \ln d, \tag{26}$$

exactly the dynamic range of the internal Hilbert space. Saturation therefore has a precise information-theoretic meaning: every Bremermann sample receives exactly one write operation carrying $\ln d$ nats, the SNR equals the dynamic range, and the Bremermann channel capacity C is exactly utilized sample by sample. The horizon is the unique surface at which the UL noise-floor condition, the Newtonian interaction rate, and the Bremermann channel capacity coincide in a single-nat write event.

For composite systems the internal Hilbert space includes additional degrees of freedom beyond spin and helicity; in general C is an upper bound on the processing rate rather than a guaranteed saturation rate. The tick-tock construction of Appendix A provides a concrete case where each Bremermann sample accesses the full two-dimensional Weyl spinor space via the Higgs vertex, saturating the dynamic range at $d = 2$. Whether this extends to more complex internal spaces is left to future work.

4.4 Relation to Existing Literature

The Page–Wootters mechanism [20, 21] derives time evolution relationally: a global static state satisfying a Wheeler–DeWitt-like constraint yields emergent dynamics when conditioned on clock eigenstates $|t\rangle$, but leaves the physical identity of the clock unspecified. The present framework supplies that identity: the matter substrate is the clock, with phase $e^{iS/\hbar}$ advancing at the Bremermann rate mc^2/h, and each communication unit acting as a PVM conditioning step that advances the Bremermann count by one quantum of action.

The modular flow perspective of Connes and Rovelli [22] identifies the modular flow σ_t^ω generated by a state ω on a von Neumann algebra as a canonical thermodynamic time. The Bisognano–Wichmann theorem identifies this flow concretely with Rindler time evolution restricted to one wedge. The present framework adds a discrete substrate beneath that flow: the continuous modular parameter is the thermodynamic limit of discrete Bremermann samples, with the horizon $\mathcal{H}$ as the boundary of the observable subalgebra that generates the modular flow as a consequence.

Verlinde's entropic gravity [2] operates at the thermodynamic level of the same horizon structure; the present framework supplies the microstate layer beneath it, with each communication realising one write operation into matter memory carrying $\ln d$ nats, incrementing the horizon area by $\Delta A = 8\pi G\hbar\omega$, and with the interaction rate $\dot{N}(a)$ recovering the Unruh thermal flux in bulk at saturation.

The framework is validated from two independent directions within this paper: the saturation result of Section 4.3 and the event-by-event GHY equality of Section 5.

5 pp-Wave Geometry and Action

We work in the null coordinates u, v defined in Section 3, with $\mathcal{P}$ at $(u, v) = (0, 0)$, $\mathcal{R}$ at $(0, 2)$, and the photon propagating along $u = 0$ and intersecting the past right Rindler horizon at $(0, 1)$; the left wedge is symmetric.

5.1 The AS pp-Wave and Its Effects

Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by

$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{27}$$

   This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{28}$$

   directed toward the axis. Test particles are pulled toward the center of the wave, with the kick strength growing as $1/r$ toward the axis.

5.2 The Longitudinal Shift Expands Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by $\mathcal{P}$ propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $+u$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction. See Figure 5.

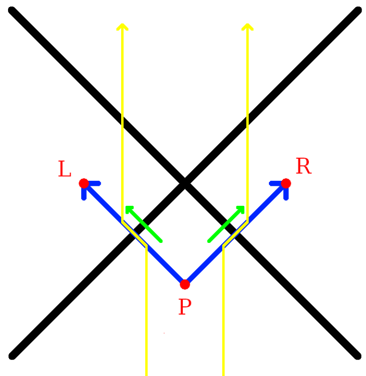

Figure 5: The green arrows show the longitudinal shift that occurs where the photons cross the horizon. Test particles, shown in yellow, are pushed outwards away from $z = 0$ due to the expansion of space along the null lines originating from $\mathcal{P}$.

At the intersection of the shock front $u = 0$ with the past right Rindler horizon at $(u, v) = (0, 1)$, the transverse (x, y) plane is identified with the horizon area elements $dA = dx\,dy$. The longitudinal shift $\Delta v \propto -\ln r^2$ displaces the horizon outward in the

v direction, pushing the right wedge geometry away from the horizon. The entangled partner photon produces a symmetric shift in the u direction at the left Rindler horizon, pushing the left wedge away from its horizon. The two wedges are therefore displaced away from their respective horizons in opposite null directions.

The geometry of this space expansion has a clean local picture. The photon travels along a null line, and the AS pp-wave it sources is mostly supported in a cylinder around that trajectory, with cross-sectional radius set by the transverse profile $\sigma_\perp$ and shrinking to the photon wavelength $\sim c/\omega$ in the localized limit. The cut-and-paste surgery of the pp-wave is performed along this null cylinder: the geometry inside is displaced in the $+v$ direction relative to the geometry outside, and it is this relative displacement that constitutes the expansion of new coordinate space. The horizon area increment, equation (31), is the integrated effect of this surgery over the cross-section of the cylinder, not a global rearrangement of the geometry but a local increment, patch by patch, concentrated around the photon trajectory. In the localized limit the patch shrinks to a Planck-scale tube and the surgery is maximally local; in the delocalized limit it spreads uniformly across the horizon. Either way the total expanded space is the same, fixed by the photon energy alone. [9]

5.3 Transverse Focusing Increments Horizon Area

The area increment ΔA is independent of the transverse localization of the photon: the Raychaudhuri equation depends only on the total null energy $\int T_{uu}\,du\,d^2x_\perp = \hbar\omega$, fixed by energy conservation independently of the transverse profile $\sigma_\perp$. We work in the localized limit $\sigma_\perp \to 0$, which recovers the AS pp-wave; the electromagnetic stress-energy T_{uu} is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane, and recovers the strict AS limit as $\omega \to \infty$ with the logarithmic divergence regulated at the scale c/ω.

We derive the area increment using the Raychaudhuri equation on the null congruence generating the Rindler horizon, following Jacobson [1] and consistent with the generalized second law for Rindler horizons proved by Wall [23], applied here to a single microstate rather than a thermodynamic or ensemble average.

For a null congruence with generators k^μ and expansion θ, the Raychaudhuri equation gives

$$\frac{d\theta}{du} = -\tfrac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - 8\pi G\, T_{\mu\nu}k^\mu k^\nu. \tag{29}$$

Integrating over the horizon patch in the linearised regime and using $k^\mu = \delta^\mu_u$, the area change is

$$\Delta A = 8\pi G \int T_{uu}\,du\,d^2x_\perp. \tag{30}$$

The present construction provides a natural regularization of the strict AS point-source limit: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth transverse profile set by the wavelength $\sim c/\omega$.

The electromagnetic stress-energy T_{uu} is well-defined throughout and integrates to $\hbar\omega$ over the transverse plane: this follows from the bilocal Lagrangian of Section 3.5, where smearing f_R with A_i^+ projects onto the Unruh mode μ_R, encoding the deposition of the full photon energy $\hbar\omega$ in the right wedge as a consequence of the projection structure rather than an assumption about absorption. The strict AS limit is recovered as $\omega \to \infty$ with the logarithmic divergence regulated at scale $\sigma > 0$.

Substituting into equation (30) gives

$$\boxed{\Delta A = 8\pi G\,\hbar\omega = 8\pi G\,\Delta E.} \tag{31}$$

[9]The physical picture is that Bremermann samples accumulated along the worldline of $\mathcal{P}$ are timelike; the null propagation and AS shift convert them into spacelike separation between the wedges, stretching proper time at $\mathcal{P}$ into new coordinate space along the null lines. The spatial bit of Section 6 is the informational record of this conversion.

The horizon area grows by one discrete, calculable increment per photon absorbed, microstate by microstate, without reference to bulk thermodynamic averages. This is property (iv) of the communication definition of Section 4.2 verified geometrically: the area increment $\Delta A = 8\pi G\hbar\omega$ is the geometric expression of one quantum of action $\hbar$ deposited into matter memory by one measurement event, with the Planck area ℓ_P^2 as the conversion factor between the two descriptions.

Here the null generators of the horizon are taken to be affinely parametrised as $k^\mu = (\partial/\partial u)^\mu$ in double-null coordinates, so that $\int T_{uu}\, du\, d^2x_\perp = \hbar\omega$ is the total null energy of the photon. This differs from the Killing-normalised convention by a factor of the surface gravity $\kappa = a$; the two are related by $(\partial/\partial u)^\mu = a(\partial/\partial\eta)^\mu$ at the horizon, consistent with the standard result $\Delta A = 8\pi GE/\kappa$ when $E = \hbar\omega$ is the Killing energy.

The longitudinal shift $\Delta v \propto -\ln r^2$ produced by each pp-wave at the Rindler horizon is a supertranslation in the sense of Bondi–Metzner–Sachs (BMS) symmetry[10]. Hawking, Perry, and Strominger [24] showed explicitly that a black hole is supertranslated by the passage of an asymmetric shockwave, and the AS longitudinal shift is precisely the linearised supertranslation field on the horizon. Carlip [25] demonstrated that the near-horizon diffeomorphisms of a generic black hole are enhanced to a BMS_3 algebra whose central charge determines the Bekenstein–Hawking entropy, but Carlip's program identifies the symmetry without specifying the physical process that excites individual modes. The bilocal construction supplies that process: each measurement event produces one AS pp-wave and deposits one supertranslation at the Rindler horizon. The shockwave S-matrix program of 't Hooft [26] established that the longitudinal AS shift is the fundamental dynamical ingredient for horizon information exchange at the level of the full S-matrix; the present construction realizes the same shift at the level of a single microstate.

5.4 Energy Conservation in Two Languages

We show how the equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ follows from energy conservation applied twice to the same physical events: once on the matter side, where each absorbed photon contributes $\hbar\omega_i$ to the null-surface action, and once on the geometric side, where the same energy drives the Raychaudhuri equation and increments the horizon area by $\Delta A_i = 8\pi G\hbar\omega_i$. The Planck area ℓ_P^2 is the conversion factor between the two descriptions, and the equality is its expression. That it holds event by event, rather than only in thermodynamic average, is the content of the calculation.

The thermodynamic interpretation of the GHY boundary term on null surfaces is established by Chakraborty and Padmanabhan [27]; the broader program of Padmanabhan [28] identifies gravitational boundary terms as the fundamental degrees of freedom at the ensemble level. The present construction supplies the microstate layer beneath those programs: the physical process underlying each boundary contribution is energy conservation at a single communication event.

Each absorption event transfers null energy $\hbar\omega_i$ across the horizon. The matter action at the horizon is defined as the null-surface integral of the stress-energy, evaluated over the horizon generators:

$$S_{\text{matter}} = \sum_i \int T_{uu}^{(i)}\, du\, d^2x_\perp = \sum_i \hbar\omega_i, \tag{32}$$

where the second equality follows from energy conservation applied to each photon. This is the null-surface analog of the worldline action $\int E\, d\tau$: on a null surface the affine parameter u plays the role that proper time plays on a timelike worldline, and the integral $\int T_{uu}\, du$ plays the role of $\int E\, d\tau$.

[10]The BMS mode content excited by different transverse profiles for our microstate, and the physical selection principle for $\sigma_\perp$, are left to future work.

The two are not the same integral. After the absorption event, the detector's worldline action continues to accumulate as $\hbar\omega_i\,\Delta\tau$ for all subsequent proper time $\Delta\tau$, encoding the Bremermann memory of the event. The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ holds for the null-surface version of the matter action, evaluated at the moment of horizon crossing; it is a statement about the communication itself, not about the subsequent worldline evolution. The Rindler proper time η and the affine null parameter u are related by $u \propto e^{-a\eta}$, and it is this exponential relation that is responsible for the Unruh temperature and that distinguishes the null-surface action from the worldline action. This is the matter action counted sample by sample: one quantum of action per event, one event per emitted pair.

The same events have a geometric description. By eq. (31), each absorption sample contributes a horizon area increment $\Delta A_i = 8\pi G\hbar\omega_i$, so the total area increment summed over all events is

$$\sum_i \Delta A_i = 8\pi G \sum_i \hbar\omega_i = 8\pi G\, S_{\text{matter}}. \tag{33}$$

This geometric sum has a precise identification in the gravitational action. On a stationary Rindler horizon the unperturbed expansion of the null generators vanishes, $\theta = 0$, so the Gibbons–Hawking–York boundary term [29] is sourced entirely by the AS pp-wave perturbations:

$$S_{GHY}\big|_{\mathcal{H}} = \frac{1}{8\pi G}\int_{\mathcal{H}} \theta\, d^2x_\perp\, du = \frac{1}{8\pi G}\sum_i \Delta A_i = \frac{1}{8\pi G}\cdot 8\pi G \sum_i \hbar\omega_i = \sum_i \hbar\omega_i = S_{\text{matter}}. \tag{34}$$

The ledger closes without remainder, as consistency requires. Chakraborty and Padmanabhan [27] established $S_{GHY}|_{\mathcal{H}} =$ heat content at the level of thermodynamic averages; the present calculation is the microstate-level version of the same accounting. The Raychaudhuri equation and the communication definition of Section 4.2 are two languages for the same events, and the fact that they agree is the check.

This is consistent with Jacobson [1] and Padmanabhan [28], who argue that the gravitational boundary term is the fundamental gravitational degree of freedom. Those programs work at the level of thermodynamic averages; the present construction works event by event. The ensemble average recovers their results; the microstate picture supplies what those programs leave open. The equality holds because it must: it is energy conservation written in two languages simultaneously, with the Planck area as the dictionary between them.

6 Entanglement as Spatial Information

The bilocal construction produces a concrete object that did not exist in the earlier sections: a classical bit that simultaneously encodes a measurement outcome and a location in space. The vacuum decomposition (16) is precisely the thermofield double state of the two Rindler wedges, now in flat Minkowski space with helicity labels carried through. Van Raamsdonk [30] showed that entanglement entropy between two halves of a thermofield double controls the geometry connecting them; the bilocal construction realizes this at the microstate level: each measurement event produces a single AS longitudinal shift displacing the two wedges apart in opposite null directions, expanding coordinate space between them. The entanglement is not a property of the geometry; it is the *cause* of it, event by event.[11]

The bilocal source prepares an anticorrelated Bell state (18). Both $\mathcal{R}$ and $\mathcal{L}$ measure in the same basis and obtain definite outcomes, which we label H_+ or H_-, perfectly

[11] In the AdS setting this is the ER=EPR correspondence [31], with the ER bridge playing the role of the AS longitudinal shift. The present construction realizes the same identification in flat Minkowski space without holographic prerequisites; the precise relationship between the AS shift and the ER bridge is left to future work.

anticorrelated by the Bell state structure independently of which basis is chosen. The question

"who observed H_+?"

has a binary answer:

$$\mathcal{L} \text{ (left wedge) or } \mathcal{R} \text{ (right wedge).} \tag{35}$$

This is a bit of information that converts a polarization state into a location in space, a measurement in both geometric and informational contexts simultaneously.

The spatial bit is the precise sense in which thermality and spatial information are the same phenomenon. Restricting to either wedge alone traces out the other; the reduced state $\rho_{\mathcal{R}} = \mathrm{Tr}_{\mathcal{L}}|\Psi_\omega\rangle\langle\Psi_\omega| = \frac{1}{2}(|1_{\omega,+1}\rangle\langle 1_{\omega,+1}| + |1_{\omega,-1}\rangle\langle 1_{\omega,-1}|)$ is maximally mixed over helicities, exactly thermal, with no preferred outcome and no spatial information, precisely as in the vacuum decomposition (16) and Bogoliubov transformation. The spatial bit is the off-diagonal correlator that this trace destroys, invisible from within either wedge alone and exactly recoverable by a future observer $\mathcal{F}$ with access to both records. The Unruh effect is therefore what the spatial bit looks like from inside one wedge: the thermality of the Rindler vacuum is the inaccessibility of the $\mathcal{R}/\mathcal{L}$ distinction from within either wedge alone. Space, in this construction, is indexed by helicity, and the spatial bit is the record of that indexing written into matter memory at each horizon crossing. This construction gives you a way to define a region of space in terms of a measurement event and its informational content.

7 Conclusion

The standard picture of spacetime puts geometry first: matter lives in space, moves through time, and gravity tells it how. This paper inverts that priority. Matter is the substrate: proper time is action accumulated along worldlines, space is the record of communication units at horizons, and the Planck area ℓ_P^2 is the conversion factor between the two descriptions. Geometry is not the stage on which physics happens; it is the ledger in which matter's interactions are recorded.

The horizon is where this inversion becomes sharp. As a causal surface it is ontic: signals cannot cross from $\mathcal{L}$ to $\mathcal{R}$. As an epistemic boundary it is the surface at which a single observer's description becomes irreducibly incomplete, not because the physics is hidden but because the information is held by someone else. The Bekenstein–Hawking entropy is the measure of that gap, and the Planck area is its conversion factor into geometry. Thermality is not a primitive feature of the vacuum; it is what remains when the positions of $\mathcal{P}$ and $\mathcal{R}$ are unspecified and the other observer's record is inaccessible. Specify both, and the thermal ensemble resolves into individual non-thermal microstates, each a single measurement event, each contributing one Planck-area patch to the horizon.

The deepest consequence is the spatial bit. The same event that records *what* was observed also determines *where*: the two wedges carry opposite helicities by construction, so the polarization outcome and the location are the same classical fact written into matter memory. What the construction shows is that the vocabulary needed to pose these questions, spatial bits, communications, matter memory, PVM projections, the Planck area as conversion factor, is the same vocabulary in which the answer, when it comes, will have to be written. Space, in this construction, is indexed by helicity, and that indexing, event by event, is what the geometry records. This construction provides an operational definition of a region of space in terms of a measurement event and its informational content.

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A The Tick-Tock Mechanism

The communication interactions from Section 4.2 have a natural analogue along a timelike worldline that occurs inevitably for massive particles in QFT. Dirac's two-component formulation of the electron [32] represents a massive fermion as two coupled Weyl spinors of opposite helicity; Penrose [33] recast this geometrically as a zig-zag of massless segments alternating in helicity, coupled at each vertex by the Higgs field at the Compton scale $\lambda_C = \hbar/mc$. The construction is a direct consequence of the two-component formalism for massive fermions. We term this the tick-tock mechanism to emphasise the alternating rhythm of communications rather than the geometry of the path.

For the Dirac field, the tick-tock structure is not an analogy but a consequence of the two-component formalism: the Higgs coupling of opposite-helicity Weyl spinors at the Compton scale $\lambda_C = \hbar/mc$ is the timelike counterpart of the bilocal construction, with the Higgs vertex playing the role of $\mathcal{P}$, the two Weyl spinors playing the role of the two photons propagating to $\mathcal{R}$ and $\mathcal{L}$, and each vertex depositing one quantum of action and producing one sample of proper time rather than one increment of horizon area. The Compton scale connection is not incidental: the Bremermann sampling rate $f = mc^2/h$ of Section 4.1 and the Compton wavelength satisfy $\lambda_C \cdot f = c$, linking the timelike sampling rate directly to the scale of the zig-zag vertices. Whether this correspondence extends beyond the Dirac field, in particular, whether the Higgs coupling vertex and the photon emission vertex of Section 3 are instances of the same communication unit in the sense of Section 4.2, is left to future work.

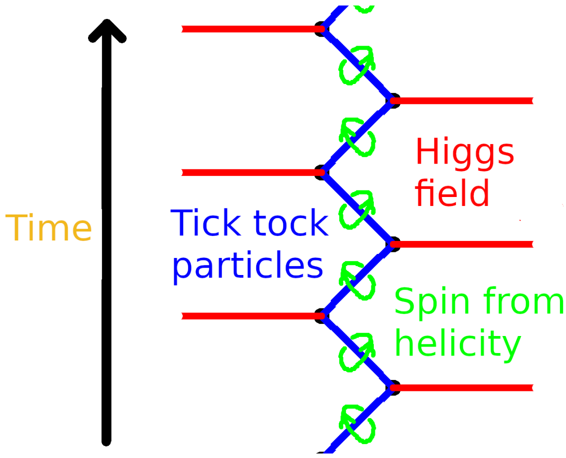

Figure 6: The tick-tock mechanism. A massive particle (blue) alternates between right- and left-helicity massless segments, coupled at each vertex by the Higgs field (red). Spin emerges from the alternating helicity flip at each vertex (green loops), visible as the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$ cycling between communications. Each vertex is a communication event: one quantum of action deposited, one helicity outcome recorded, one sample of proper time produced.

The anticorrelated helicity assignment in both constructions has a common algebraic origin. The Bisognano–Wichmann modular conjugation J acts on right-wedge photon creation operators as

$$J\, a^{R\dagger}_{k,\lambda}\, J^{-1} = a^{L\dagger}_{-k,-\lambda}, \tag{36}$$

flipping helicity and reversing the spacelike separation direction $R \leftrightarrow L$. The PCT operator Θ acts on a right-handed Weyl spinor as

$$\Theta\, \psi_R(t, \mathbf{x})\, \Theta^{-1} \propto \psi_L(-t, -\mathbf{x}), \tag{37}$$

flipping helicity and reversing the timelike separation direction $t \to -t$. In both cases the operation is the same at the representation-theoretic level: exchange of the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$. The operators J and Θ act on different Hilbert spaces; what they share is that both select anticorrelated helicities as the unique PCT-invariant configuration.

B Verlinde-style Heuristics

The Landauer–Unruh bound of Section 2.1 and the momentum transfer of Section 2.2 motivate a set of Newtonian heuristics that recover standard black hole thermodynamics without invoking temperature. The heuristic input is the identification of a measurement event localized to spatial scale r with a causal diamond of temporal extent $\Delta t = 2r/c$, giving

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2 m_d r}. \tag{38}$$

From this single identification, combined with Newton's laws and $E = mc^2$, standard results follow as heuristic consequences. These derivations follow Verlinde [2] in using dimensional analysis at the Schwarzschild scale; the coincidence exploited is that the Newtonian inverse-square law reproduces the Schwarzschild geometry at the horizon scale in the spherically symmetric case. The construction of Sections 3–5 makes no use of these results.

B.1 The Schwarzschild Radius

We match the kinematic acceleration of the detector to the gravitational field of the thrust quantum using Newton's laws. By $E = mc^2$, the energy quantum ΔE gravitates with effective mass $\Delta E/c^2$, producing a Newtonian gravitational acceleration

$$a = \frac{G \Delta E}{c^2 r^2} \tag{39}$$

at distance r from the detector. Equating this with the measurement-induced acceleration from equation (38),

$$\frac{\Delta E}{2 m_d r} = \frac{G \Delta E}{c^2 r^2}. \tag{40}$$

The energy ΔE of the thrust quantum cancels entirely: the Schwarzschild radius is independent of the energy of the carrier. Solving for r:

$$r = \frac{2 G m_d}{c^2} \equiv r_s, \tag{41}$$

which is precisely the Schwarzschild radius of the detector mass m_d. The cancellation of ΔE is the physical content of the result: whatever energy the thrust quantum carries, the self-consistent horizon scale is set by the detector alone.

B.2 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{Gm_d}{r_s^2} = \frac{c^4}{4Gm_d}, \tag{42}$$

in agreement with the standard Schwarzschild result, as expected from the Verlinde coincidence. This reproduces the acceleration appearing in the Unruh temperature $T = \hbar\kappa/2\pi k_B c$, providing an internal consistency check on the heuristics.

B.3 Bekenstein–Hawking Entropy

Returning to the Landauer–Unruh relation (3) and substituting the surface gravity κ for a, justified by the equivalence principle: the local quantum field theory at the black hole horizon is the same as in the Rindler frame [11, 34], the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta H \leq \frac{2\pi c}{\hbar\kappa}\Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot \Delta m_d c^2. \tag{43}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta H \leq \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{44}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

This derivation parallels Jacob Bekenstein's original argument [35], in which the entropy increase of a black hole is inferred from the minimal energy required to localize and drop a system across the horizon; here, the same structure emerges from the the minimal carrier energy required to register one distinguishable event at the horizon scale.

The two bounds are saturated simultaneously when $a = c^2/R$, which is exactly the condition for being *at a horizon*. The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the Landauer dissipation bound and the Bekenstein entropy bound coincide.

Substituting the horizon condition $a = c^2/R$ directly into the Landauer–Unruh bound (3) yields

$$\Delta H \leq \frac{2\pi R\, \Delta E}{\hbar c}, \tag{45}$$

which is precisely the Bekenstein universal entropy bound [36] on the entropy-to-energy ratio for a system of size R. The Landauer–Unruh relation and the Bekenstein bound are therefore the same inequality: the former is its acceleration-frame expression, the latter its position-space expression, and the Rindler horizon is the surface at which the two descriptions coincide. [12]

Pesci [38] derived the covariant entropy bound from the Unruh temperature via the second law, the closest precursor to the present argument. The present bound (3) differs in that temperature is eliminated entirely and the bound is applied to individual carrier quanta rather than a bulk thermodynamic flux. The identification of equation (3) with the Bekenstein bound [36] at the horizon condition $a = c^2/R$ as the same inequality in different frames appears to be new in this form.

[12] The Bousso bound [37] generalizes the Bekenstein bound to arbitrary null congruences; the present construction can be read as a microstate-level realisation, with each communication contributing one Planck-area patch to the light-sheet.

References

[1] T. Jacobson, "Thermodynamics of spacetime: the einstein equation of state," *Physical review letters*, vol. 75, no. 7, p. 1260, 1995.

[2] E. Verlinde, "On the origin of gravity and the laws of newton," *Journal of High Energy Physics*, vol. 2011, no. 4, pp. 1–27, 2011.

[3] K. Player, "Driving the unruh response," *arXiv preprint arXiv:2509.16710*, 2025.

[4] L. Diosi, "A universal master equation for the gravitational violation of quantum mechanics," *Physics letters A*, vol. 120, no. 8, pp. 377–381, 1987.

[5] R. Penrose, "On gravity's role in quantum state reduction," *General relativity and gravitation*, vol. 28, no. 5, pp. 581–600, 1996.

[6] S. A. Fulling, "Nonuniqueness of canonical field quantization in riemannian spacetime," *Physical Review D*, vol. 7, no. 10, p. 2850, 1973.

[7] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a hermitian scalar field," *Journal of Mathematical Physics*, vol. 16, no. 4, pp. 985–1007, 1975.

[8] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM journal of research and development*, vol. 5, no. 3, pp. 183–191, 1961.

[9] M. Cortês and A. R. Liddle, "Hawking evaporation and the landauer principle," *arXiv preprint arXiv:2407.08777*, 2024.

[10] Y. J. Alvim and L. C. Céleri, "Landauer principle and the second law in a relativistic communication scenario," *Entropy*, vol. 26, no. 7, p. 613, 2024.

[11] W. G. Unruh, "Notes on black-hole evaporation," *Physical Review D*, vol. 14, no. 4, p. 870, 1976.

[12] S. Hawking and W. Israel, *General Relativity: An Einstein Centenary Survey.* Cambridge University Press, 1979.

[13] N. Harrigan and R. W. Spekkens, "Einstein, incompleteness, and the epistemic view of quantum states," *Foundations of Physics*, vol. 40, no. 2, pp. 125–157, 2010.

[14] C. C. Gerry and P. L. Knight, *Introductory quantum optics.* Cambridge university press, 2023.

[15] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space.* Cambridge: Cambridge University Press, 1984.

[16] J. A. Wheeler, "Geons," *Physical Review*, vol. 97, no. 2, p. 511, 1955.

[17] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-time as a causal set," *Physical review letters*, vol. 59, no. 5, p. 521, 1987.

[18] K. Brown, "Stochastic sprinkling." MathPages. Accessed: 2026-04-19.

[19] H. J. Bremermann, "Optimization through evolution and recombination.," *Self-organizing systems*, pp. 93–106, 1962.

[20] D. N. Page and W. K. Wootters, "Evolution without evolution: Dynamics described by stationary observables," *Physical Review D*, vol. 27, no. 12, p. 2885, 1983.

[21] V. Giovannetti, S. Lloyd, and L. Maccone, "Quantum time," *Physical Review D*, vol. 92, no. 4, p. 045033, 2015.

[22] A. Connes and C. Rovelli, "Von neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories," *Classical and Quantum Gravity*, vol. 11, no. 12, pp. 2899–2917, 1994.

[23] A. C. Wall, "Proof of the generalized second law for rapidly evolving rindler horizons," *Physical Review D—Particles, Fields, Gravitation, and Cosmology*, vol. 82, no. 12, p. 124019, 2010.

[24] S. W. Hawking, M. J. Perry, and A. Strominger, "Soft hair on black holes," *Physical Review Letters*, vol. 116, no. 23, p. 231301, 2016.

[25] S. Carlip, "Black hole entropy from bondi-metzner-sachs symmetry at the horizon," *Physical review letters*, vol. 120, no. 10, p. 101301, 2018.

[26] G. Hooft, "The scattering matrix approach for the quantum black hole: an overview," *International Journal of Modern Physics A*, vol. 11, no. 26, pp. 4623–4688, 1996.

[27] S. Chakraborty and T. Padmanabhan, "Boundary term in the gravitational action is the heat content of the null surfaces," *Physical Review D*, vol. 101, no. 6, p. 064023, 2020.

[28] T. Padmanabhan, "Thermodynamical aspects of gravity: new insights," *Reports on Progress in Physics*, vol. 73, no. 4, p. 046901, 2010.

[29] G. W. Gibbons and S. W. Hawking, "Action integrals and partition functions in quantum gravity," *Physical Review D*, vol. 15, no. 10, p. 2752, 1977.

[30] M. Van Raamsdonk, "Building up space–time with quantum entanglement," *International Journal of Modern Physics D*, vol. 19, no. 14, pp. 2429–2435, 2010.

[31] J. Maldacena and L. Susskind, "Cool horizons for entangled black holes," *Fortschritte der Physik*, vol. 61, no. 9, pp. 781–811, 2013.

[32] P. A. M. Dirac, "The quantum theory of the electron," *Proceedings of the royal society of London. Series A, Containing papers of a mathematical and physical character*, vol. 117, no. 778, pp. 610–624, 1928.

[33] R. Penrose, *The road to reality.* Random house, 2006.

[34] S. W. Hawking, "Particle creation by black holes," *Communications in mathematical physics*, vol. 43, no. 3, pp. 199–220, 1975.

[35] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D*, vol. 7, pp. 2333–2346, Apr 1973.

[36] J. D. Bekenstein, "Universal upper bound on the entropy-to-energy ratio for bounded systems," *Physical Review D*, vol. 23, no. 2, p. 287, 1981.

[37] R. Bousso, "A covariant entropy conjecture," *Journal of High Energy Physics*, vol. 1999, no. 07, pp. 004–004, 1999.

[38] A. Pesci, "From unruh temperature to the generalized bousso bound," *Classical and Quantum Gravity*, vol. 24, no. 24, pp. 6219–6225, 2007.